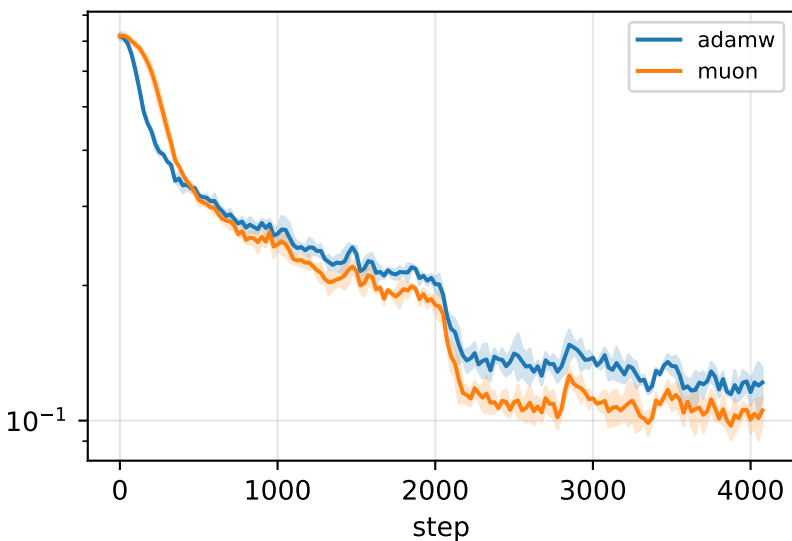
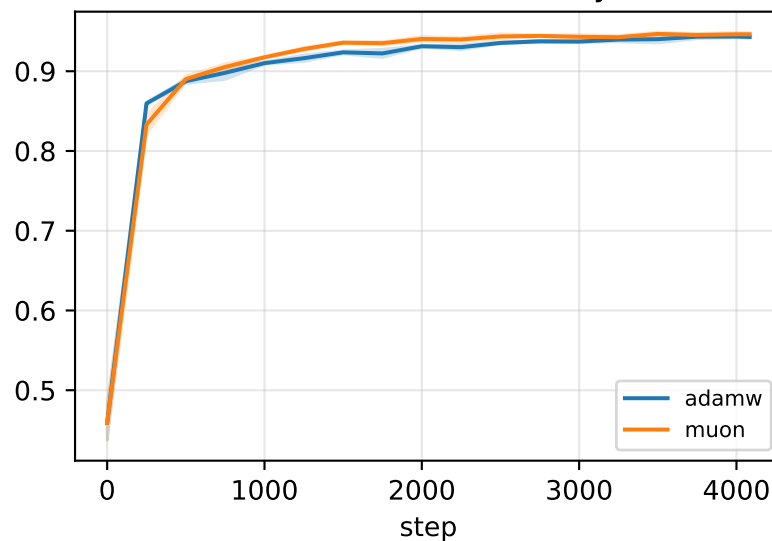


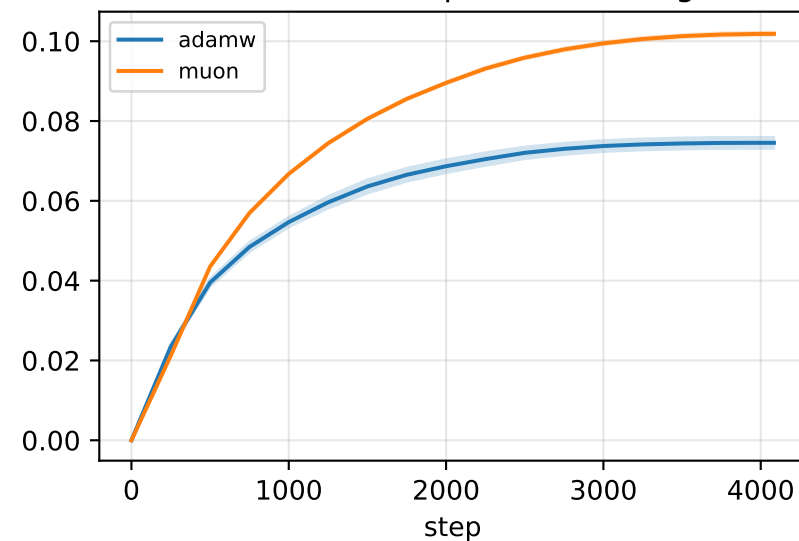
(a) train loss



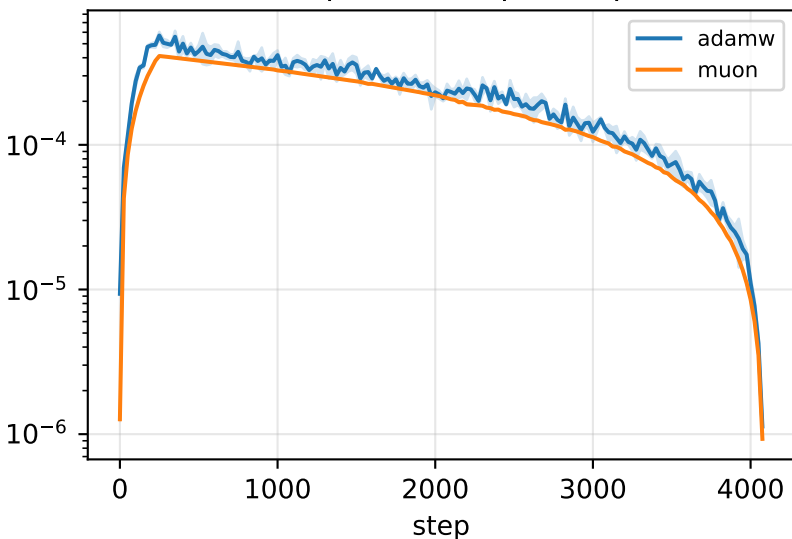
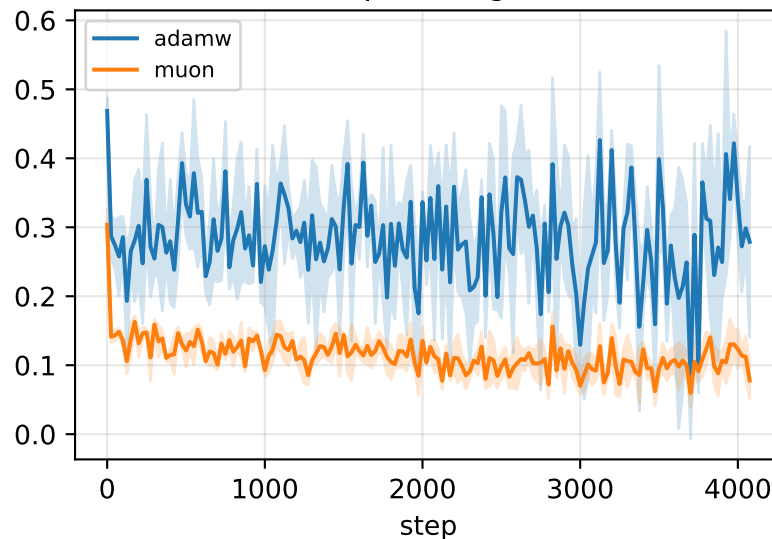
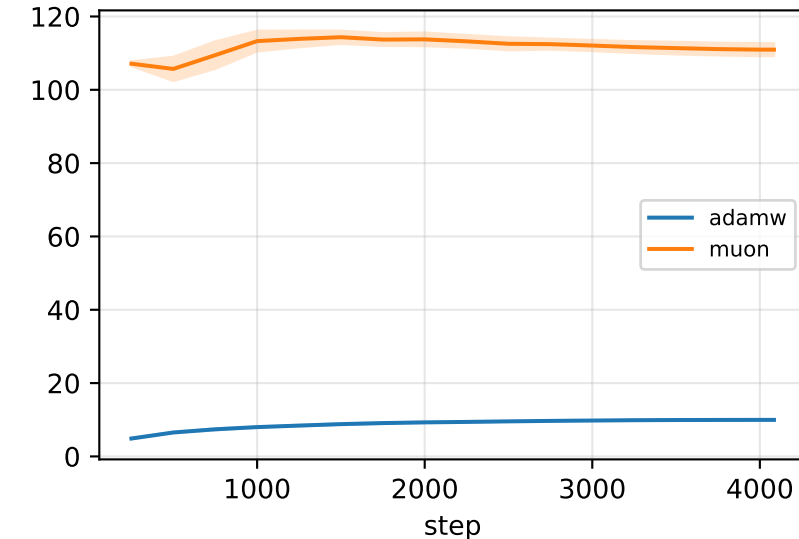
(b) validation accuracy



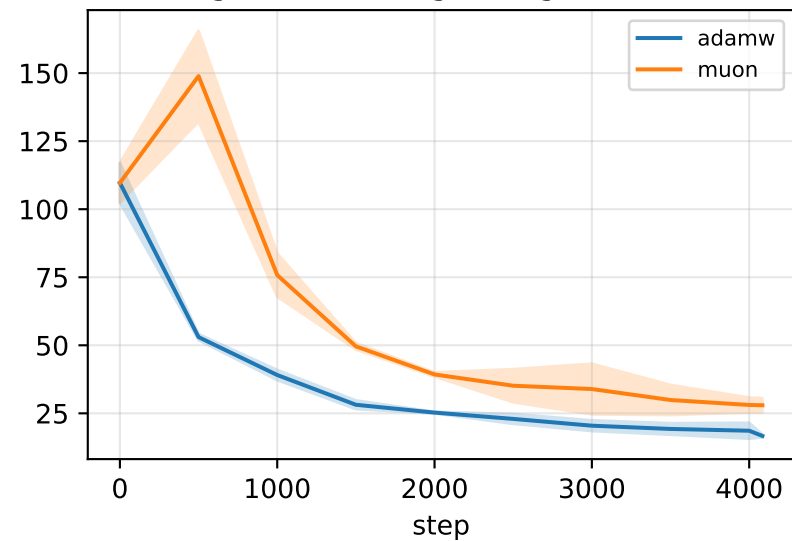
(c) distance from pretrained weights



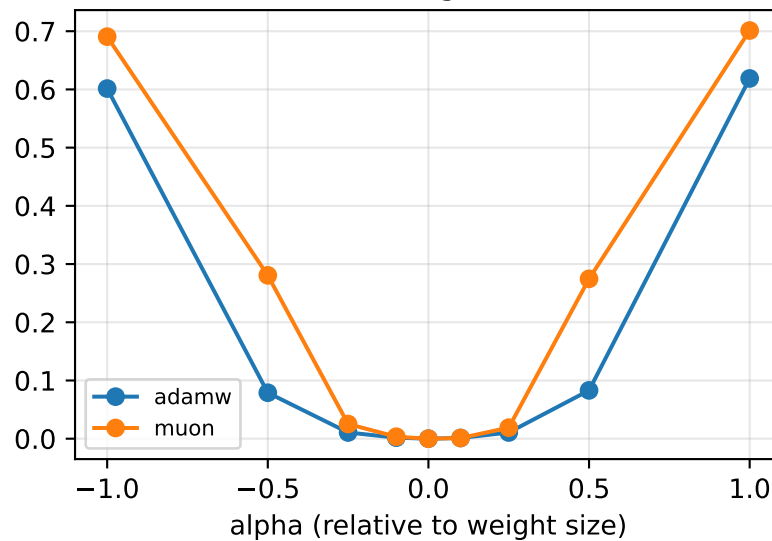
(d) update size per step

(e) $\cos(\text{update}, -\text{gradient})$ (f) stable rank of $W - W_0$ 

(g) Hessian largest eigenvalue



(h) loss increase along random directions

(i) AdamW ($t=0$) to Muon ($t=1$)
solid: train probe, dashed: test